

Semantic Theory 2026: Exercise 2 Key

Question 1

Provide the derivations (type inferencing) of each of the following sentences. Brackets indicate the combinatorics, and subscripts indicate the types of (some of) the expressions—the rest must be deduced. Underlined expressions should be treated as a single term: for example, you can treat “the father of” as *the-father-of* in (a).

- a. Darth Vader_e [is_{⟨e,⟨e,t⟩⟩} [the father of [Luke]_e]]

$$\frac{\frac{is :: \langle e, \langle e, t \rangle \rangle \quad \frac{father-of :: \langle e, e \rangle \quad luke :: e}{father-of(luke) :: e}}{is(father-of(luke)) :: \langle e, t \rangle} \quad darth-vader :: e}{is(father-of(luke))(darth-vader) :: t}$$

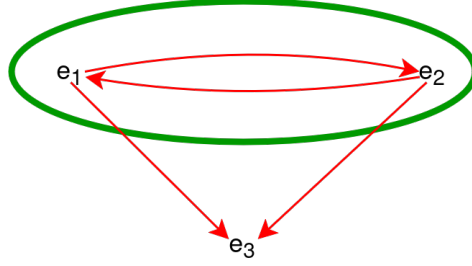
- b. [Every student_{⟨e,t⟩}] [reads the book_e]

$$\frac{\frac{every :: \langle \langle e, t \rangle, \langle \langle e, t \rangle, t \rangle \rangle \quad student :: \langle e, t \rangle}{every(student) :: \langle \langle e, t \rangle, t \rangle} \quad \frac{read :: \langle e, \langle e, t \rangle \rangle \quad the-book :: e}{read(the-book) :: \langle e, t \rangle}}{every(student)(read(the-book)) :: t}$$

- c. Mark_e [is [the_{⟨⟨e,t⟩,e⟩} [[most annoying] man_{⟨e,t⟩}]]] [on_{⟨e,⟨⟨e,t⟩,⟨e,t⟩⟩} the planet_e]

(ι = “the”)

$$\frac{\frac{\frac{on :: \langle e, \langle et, et \rangle \rangle \quad p' :: e}{on(p') :: \langle et, et \rangle} \quad \frac{is :: \langle e, et \rangle \quad \frac{\frac{\frac{most :: \langle \langle et, et \rangle, \langle et, et \rangle \rangle \quad a' : \langle et, et \rangle}{most(a') :: \langle et, et \rangle} \quad man :: \langle e, t \rangle}{\iota :: \langle et, e \rangle} \quad \frac{most(a')(man) :: et}{\iota(most(a')(man)) :: e}}{is(\iota(most(a')(man))) :: \langle e, t \rangle}}{on(p')(is(\iota(most(a')(man)))) :: \langle e, t \rangle} \quad m' :: e}{on(p')(is(\iota(most(a')(man))))(m') :: t}$$



Question 2

The diagram above graphically represents a model structure $M = (U_M, V_M)$ with a universe consisting of three entities: $U_M = \{e_1, e_2, e_3\}$. The interpretation function V_M describes the first-order property “tall” (indicated by the green circle) and the two-place relation “to see” (indicated by the red arrows).

2.1 Give the type-theoretic denotation of the interpretation function V_M for the following non-logical constants, using functions (rather than sets) when appropriate:

a. $tall \in \text{CON}_{\langle e, t \rangle}$

$$V_M(tall) = \{e_1, e_2\}$$

b. $see \in \text{CON}_{\langle e, \langle e, t \rangle \rangle}$

$$V_M(see) = \left\{ \begin{array}{l} e_1 \mapsto \left\{ \begin{array}{l} e_1 \mapsto 0 \\ e_2 \mapsto 1 \\ e_3 \mapsto 0 \end{array} \right. \\ e_2 \mapsto \left\{ \begin{array}{l} e_1 \mapsto 1 \\ e_2 \mapsto 0 \\ e_3 \mapsto 0 \end{array} \right. \\ e_3 \mapsto \left\{ \begin{array}{l} e_1 \mapsto 1 \\ e_2 \mapsto 1 \\ e_3 \mapsto 0 \end{array} \right. \end{array} \right.$$

2.2 Compute the type-theoretic denotations of the following expressions relative to the given model structure M and some arbitrary variable assignment g . Here, x, y are variables of type e , and F is a variable of type $\langle e, t \rangle$.

a. $\llbracket see(y) \rrbracket^{M, g[y/e_2]} = ?$

$$\begin{aligned} & \llbracket see(y) \rrbracket^{M, g[y/e_2]} \\ &= \llbracket see \rrbracket^{M, g[y/e_2]}(\llbracket y \rrbracket^{M, g[y/e_2]}) \\ &= V_M(see)(g[y/e_2](y)) \\ &= V_M(see)(e_2) \\ &= \{e_1\} \end{aligned}$$

b. $\llbracket \forall x(\neg \exists y(see(y)(x)) \rightarrow \neg tall(x)) \rrbracket^{M, g} = 1$

$$\begin{aligned} & \text{iff for all } d \in D_e: \llbracket \neg \exists y(see(y)(x)) \rightarrow \neg tall(x) \rrbracket^{M, g[x/d]} = 1 \\ & \text{i.e.: } d \in D_e: \llbracket \neg \exists y(see(y)(x)) \rrbracket^{M, g[x/d]} = 0 \text{ or } \llbracket \neg tall(x) \rrbracket^{M, g[x/d]} = 1 \\ & \quad \llbracket \neg \exists y(see(y)(x)) \rrbracket^{M, g[x/d]} = 0 \text{ iff } \llbracket \exists y(see(y)(x)) \rrbracket^{M, g[x/d]} = 1 \\ & \quad \text{iff there is some } b \in D_e \text{ s.t. } \llbracket see(y)(x) \rrbracket^{M, g[x/d][y/b]} = 1 \\ & \quad \text{iff } d \in V_M(see)(b) \\ & \quad \llbracket \neg tall(x) \rrbracket^{M, g[x/d]} = 1 \text{ iff } \llbracket tall(x) \rrbracket^{M, g[x/d]} = 0 \\ & \quad \text{iff } d \notin V_M(tall) \end{aligned}$$

applied to the model M :

$$\begin{aligned} & V_M(see)(e_3) = \{e_1, e_2\} \Rightarrow \llbracket \neg \exists y(see(y)(x)) \rrbracket^{M, g[x/e_1]} = 0 \text{ and } \llbracket \neg \exists y(see(y)(x)) \rrbracket^{M, g[x/e_2]} = 0 \\ & \quad \Rightarrow \llbracket \neg \exists y(see(y)(x)) \rightarrow \neg tall(x) \rrbracket^{M, g[x/e_1]} = 1 \\ & \quad \text{and } \llbracket \neg \exists y(see(y)(x)) \rightarrow \neg tall(x) \rrbracket^{M, g[x/e_2]} = 1 \\ & \quad e_3 \notin V_M(tall) \Rightarrow \llbracket \neg tall(x) \rrbracket^{M, g[x/e_3]} = 1 \\ & \quad \Rightarrow \llbracket \neg \exists y(see(y)(x)) \rightarrow \neg tall(x) \rrbracket^{M, g[x/e_3]} = 1 \\ & \Rightarrow \llbracket \forall x(\neg \exists y(see(y)(x)) \rightarrow \neg tall(x)) \rrbracket^{M, g} = 1 \end{aligned}$$

c. $\llbracket \forall F \exists x(F(x)) \rrbracket^{M, g} = 1$

$$\begin{aligned} & \text{iff for all } P \in D_{\langle e, t \rangle}: \llbracket \exists x(P(x)) \rrbracket^{M, g[F/P]} = 1 \\ & \quad \text{iff there is some } d \in D_e \text{ such that } \llbracket P(x) \rrbracket^{M, g[F/P][x/d]} = 1 \\ & \quad \text{iff } d \in V_M(P) \end{aligned}$$

we only have one one-place predicate in M : $tall$, and $e_1, e_2 \in V_M(tall)$
 $\Rightarrow \llbracket \forall F \exists x(F(x)) \rrbracket^{M, g} = 1$